Paradigms - Unification
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I) $p(b, b, a)$ and $p(X, Y, Z)$

$X = b$
$Y = b$
$Z = a$

$\{X = b, Y = b, Z = a\}$

$p(b, b, a) \rightarrow$ unified

II)

$p(f(a, b), c, d)$ and $p(f(X, Y), X, Z)$

$X = a$
$Y = b$
$X = c$
$Z = d$

$\{X = a, Y = b, X = c, Z = d\}$

$X = a = c \ldots$

does not unify...

III)

$p(f(a,b), a, d)$ and $p(f(X,Y), X, Z)$

$X = a$
$Y = b$
$X = a$
$Z = d$

$\{X = a, Y = b, Z = d\}$

$p(f(a,b), a, d)$

IV) $p(a, Z, Z)$ and $p(Y, Y, b)$

$Y = a$
$Y = Z$
$Z = b$

$\{Y = a, Y = Z, Z = b\}$

contradiction: Z does not equal Y, does not unify.

V.

$q(X, f(X, Y), Y)$ and $q(U, f(Z, c), Z)$

$X = U$
$X = Z$
$Y = c$
$Y = Z$

$\{X = U, X = Z, Y = c, Y = Z\}$

$\hookrightarrow \{X = c, U = c, Y = c, Z = c\}$

$q(c, f(c, c), c)$

VI)

$q(f(X, Y), Y, X)$ and $q(Z, U, U)$

$Z = f(X, Y)$
$Y = U$
$X = U$

$\{Z = f(X, Y), U = X, Y = X\}$

$q(f(X, X), X, X)$

VII) $q(X, Z, Z)$ and $q(U, U, Y)$

$X = U$
$Z = U$
$Z = Y$

$\{X = U, Z = U, Y = Z\}$

$q(X, X, Y)$

VIII)

$q(X, f(Y, g(X, a)))$ and $q(h(U, a), Z)$

$X = h(U, a)$
$Z = f(Y, g(X, a))$

$\hookrightarrow Z = f(Y, g(h(U, a), a))$

$\{X = h(U, a), Z = \quad\}$

$q(h(U, a), f(Y, g(h(U, a), a)))$